



Semester One Examination, 2024

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

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In words

Circle your teacher:

Mrs Alvaro

Mr Buckland

Ms Chua

Mrs Fraser-Jones

Miss K

Mr Koulianos

Mrs Lastari

Ms Lee

Mr Lei

Mr Luzuk

Mr Ng

Mr Safa

Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

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Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	6	6	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
7	6	
8	12	
9	12	
10	9	
11	5	
12	6	
13	6	
14	10	
15	13	
16	6	
17	7	
18	6	
Section 2 Total	98	
Weighted ($\times 0.6633$)	65%	

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Section Two: Calculator-assumed**65% (98 Marks)**

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 7**(6 marks)**

Given that $P(A) = 0.3$ and $P(B) = 0.5$,

determine the value of $P(A \cup B)$ in each of the following different situations.

(a) A and B are mutually exclusive.

(2 marks)

(b) $P(A|B) = 0.8$.

(2 marks)

(c) A and B are independent.

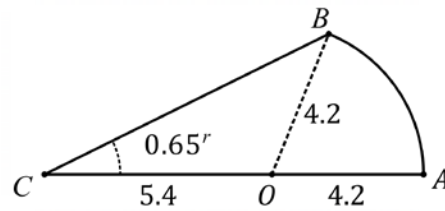
(2 marks)

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Question 8

(12 marks)

A thin metal plate ABC , as shown in the diagram (not to scale), consists of sector OAB of a circle with radius 4.2 cm and centre O joined to triangle OBC so that AOC is a straight line and $OC = 5.4$ cm.



The size of $\angle OCB = 0.65$ radians and BC is the longest side of triangle OBC .

- (a) Use the sine rule and consideration of the ambiguous case to show that the size of $\angle BOC = 1.60$ radians, correct to 2 decimal places. (4 marks)

- (b) Use the cosine rule to show that $BC = 6.94$ cm, correct to 2 decimal places. (2 marks)

(c) Determine the perimeter of the metal plate.

(3 marks)

(d) Determine the area of the metal plate.

(3 marks)

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Question 9**(12 marks)**

An ATAR Dance performance consists of 6 dancers. The dancers can be chosen from seven Year 11 students and six Year 12 students. One dancer from each year is also a choreographer.

- (a) Find the number of ways the dancers can be selected if ;
- (i) they must consist of equal numbers of Year 11s and Year 12s, (2 marks)

 - (ii) they must consist of at least one dancer from each year, (2 marks)

 - (iii) either the Year 11 choreographer or the Year 12 choreographer must be included, but not both. (2 marks)

 - (iv) there are more Year 11 dancers than Year 12 dancers. (3 marks)

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The six dancers are chosen. The choreographer splits the dancers into two groups, a and b .

(b) Expand the expression $(a + b)^6$. (1 mark)

(c) Explain what the coefficient of the a^2b^4 term means in the context of the question. (2 marks)

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Question 10

(9 marks)

- (a) Use the table of approximations shown below and an identity from the standard formula sheet to obtain an approximate value for $\cos 19^\circ$. (3 marks)

θ	$\cos \theta$	$\sin \theta$
36°	$\frac{4}{5}$	$\frac{3}{5}$
55°	$\frac{4}{7}$	$\frac{5}{6}$

- (b) Peter can see Ray in a dingy, 450 metres away on a bearing of 110° T. Mary, who is 200 metres to the south of Peter, can also see Ray in the dingy. At what bearing must Ray be looking if his gaze is fixed on Mary ? (6 marks)

Question 11**(5 marks)**

A screening test for malaria is carried out in a population in which 2% of people have malaria. The test returns one of two results - a person has the disease, or they do not. It has a sensitivity of 87% (i.e., the test indicates that people with malaria have the disease in 87% of cases) and a specificity of 94% (i.e., the test indicates that people free of malaria do not have the disease in 94% of cases).

A person is randomly chosen from the population and tested.

- (a) Determine the probability that the person has malaria and that the test indicates they have the disease. (1 mark)

- (b) Determine the probability that the person has malaria, given that the test indicates that they have malaria. (4 marks)

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Question 12**(6 marks)**

A bridge crossed a tidal inlet so that the clearance height h metres between the water surface and the bottom of the bridge varied with time t (the number of hours after low tide at 5 am) according to the rule

$$h(t) = 5.4 + 1.7 \cos\left(\frac{\pi t}{6}\right), \quad 0 \leq t \leq 24.$$

(a) Determine what the clearance height was at

(i) 5 am. (1 mark)

(ii) 8:45 am. (1 mark)

(b) At what time was the clearance first a minimum and what was the clearance at this time? (2 marks)

(c) A fishing boat required a clearance of at least 4.6 m to pass beneath the bridge. What was the latest time, to the nearest minute, that it could pass beneath the bridge in the morning? (2 marks)

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Question 13

(6 marks)

A company has placed 400 vouchers into identical unmarked envelopes as part of a promotion it is running. Each voucher admits 1, 2, 3 or 4 people to one of three sporting events as shown in the table below:

	<i>Voucher admits</i>			
	1 person	2 people	3 people	4 people
Basketball	50	35	25	40
Football	27	17	21	45
Soccer	48	20	42	30

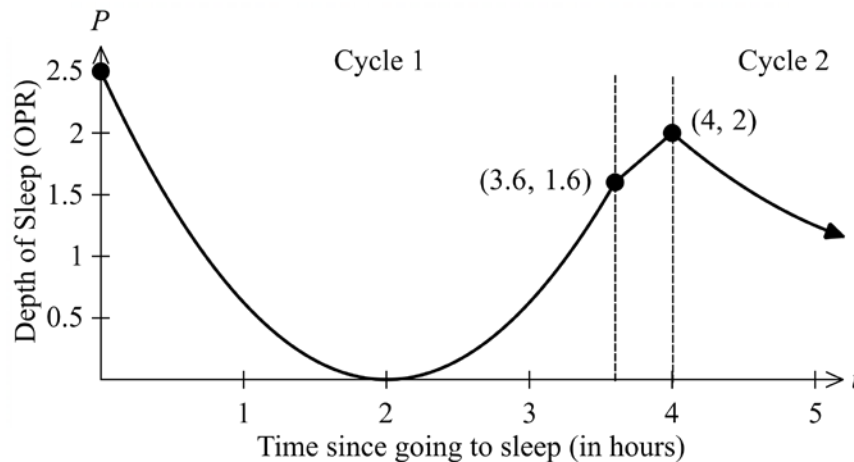
One envelope is selected at random from the 400. Determine the probability that the voucher in the envelope

- (a) is for soccer and admits 2 people. (1 mark)
- (b) is for basketball. (1 mark)
- (c) admits no more than 3 people. (1 mark)
- (d) admits one person or is for basketball. (1 mark)
- (e) admits an odd number of people given soccer is excluded (2 marks)

Question 14**(10 marks)**

The Odds Ratio Product (ORP) is used by some smart watches to measure how well a person sleeps. The ORP can range from 0 (deeply asleep) to 2.5 (fully awake). Any ORP between these two values means a person is sleeping. The lower the ORP, the greater the depth of sleep.

Josh's smart watch recorded his ORP as he slept one night. The diagram below shows the depth of his sleep, P , during the first five hours of sleep. The horizontal axis, t , is the number of hours since Josh went to sleep.



Cycle 1 shows that Josh was fully awake when he went to bed and he was deeply asleep after 2 hours.

- (a) Assuming that the graph is parabolic, determine the equation of Cycle 1. (2 marks)

The graph shows that Cycle 1 stopped after 3.6 hours. At this time, Josh's ORP was 1.6.

- (b) If Josh went to bed at 10 pm, use the line of symmetry to determine when his watch first recorded an ORP of 1.6. (2 marks)

Cycle 1 finished when $t = 3.6$ and Cycle 2 started when $t = 4$.

- (c) Determine the equation of the straight line between Cycle 1 and Cycle 2. (2 marks)

Cycle 2 started at 2 am and continued until Josh woke up.

The equation for Cycle 2 is

$$P = \frac{t^2}{4} - 3t + 10$$

- (d) Use the discriminant to show that Josh will never be deeply asleep during Cycle 2. (2 marks)

It is recommended that people get at least eight hours of sleep.

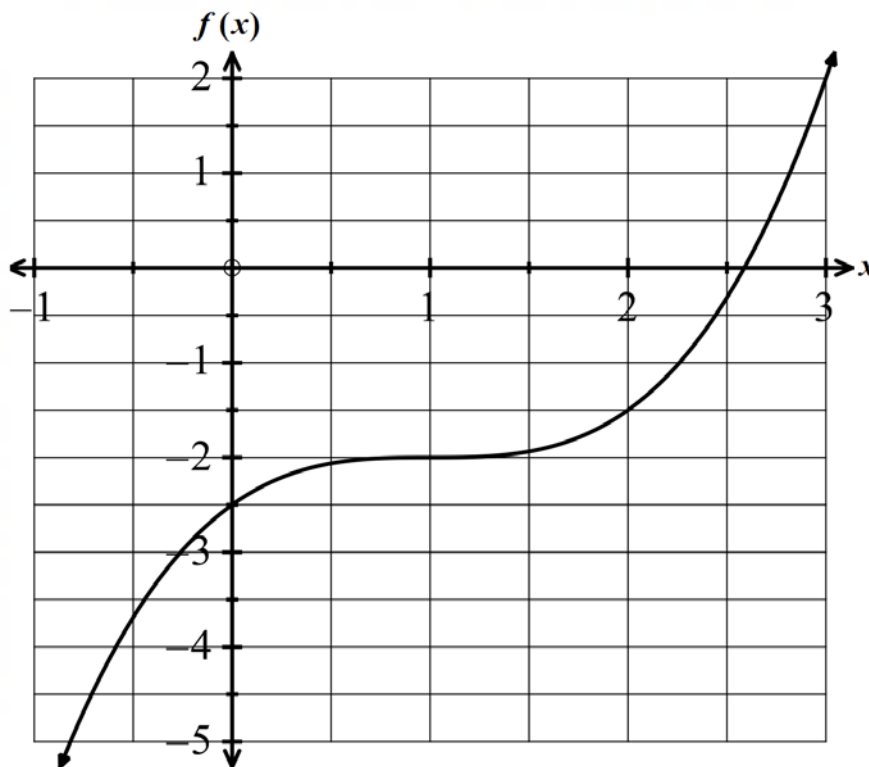
- (e) Does Josh get the recommended amount of sleep? Justify your answer mathematically. (2 marks)

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Question 15

(13 marks)

- (a) Part of the graph of $y = g(x)$, where $g(x) = a(x - b)^3 + c$ and a, b and c are constants, is shown below. The curve passes through $(0, -2.5)$ and has a horizontal point of inflection at $(1, -2)$.



- (i) Determine the value of each of the constants a, b and c . (2 marks)
- (ii) The graph of $y = g(x)$ is dilated vertically by a scale factor of 2 followed by a translation of two units to the left. Determine ;
- (α) the coordinates of the y -intercept of the transformed graph. (2 marks)
- (β) the equation of the transformed graph (2 marks)

(b) Determine the sequence of transformations required to transform

(i) $y = 1 + \frac{1}{x-2}$ to $y = \frac{1}{2x-2}$ (2 marks)

(ii) $x^2 + (y-1)^2 = 25$ to $(x+3)^2 + (y+1)^2 = 25$ (2 marks)

(c) If $f(x) = \sqrt{6-x}$ state the domain and range of $g(x) = f(2x)$ (3 marks)

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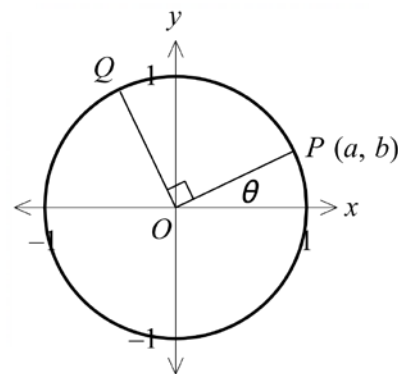
Question 16**(6 marks)**

The diagram on right shows the unit circle.

The line segment OP makes an angle of θ with the positive x axis, where $0 < \theta < \frac{\pi}{2}$.

The coordinates of P are given by (a, b) .

On the diagram, the line segment OQ is perpendicular to OP .



(a) State the value of each of the following in terms of a and/or b :

(i) $\cos(\pi - \theta)$ (1 mark)

(ii) $\tan(-\theta)$ (2 marks)

(b) (i) State the gradient of the line segment OP in terms of θ . (1 mark)

(ii) Hence, or otherwise, state the gradient of line segment OQ in terms of a and/or b . (1 mark)

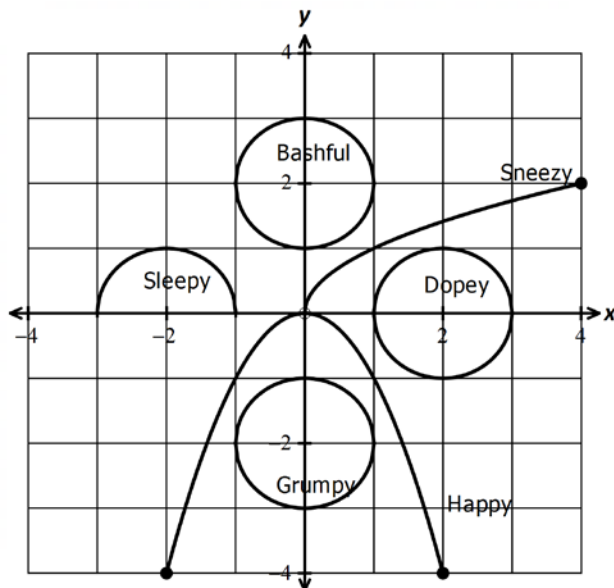
(iii) State, in terms of a and/or b , the value of $\sin\left(\theta + \frac{\pi}{2}\right)$ (1 mark)

Question 17

(7 marks)

A designer creates the 8 cm by 8 cm pattern. She names the 6 curves after her 6 relations. The relations are Sneezzy, Bashful, Sleepy, Happy, Grumpy and Dopey.

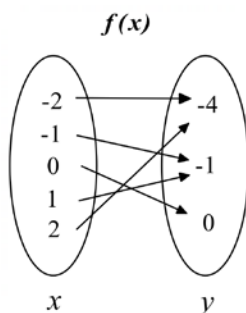
Each square is 1 cm by 1 cm, and the x and y axes are indicated.



- (a) State the equations of the curves labelled *Sneezzy*, *Grumpy* and *Happy*. (3 marks)

- (b) How many of the six relations are also functions? Explain how you made your decision. (2 marks)

Part of the mapping of one of the functions in part (b) is given below



- (c) Using the graph and the mapping, state the domain and range for this function. (2 marks)

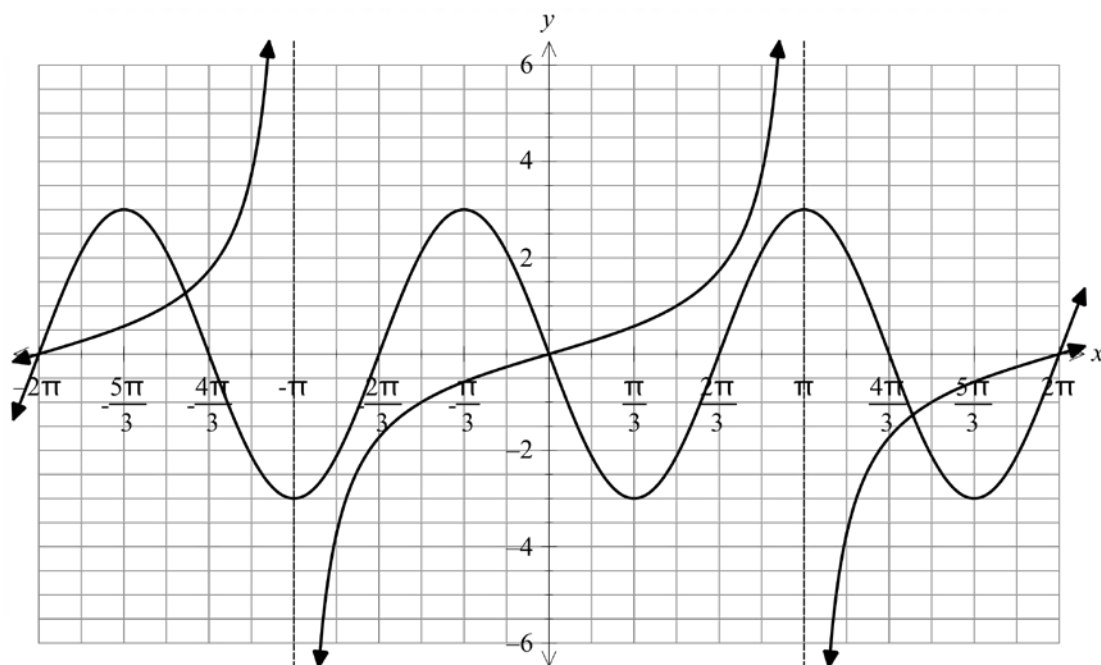
Question 18

(6 marks)

The following graphs are shown in the diagram below:

$$f(x) = a \sin bx$$

$$g(x) = \tan cx$$



- (a) Determine the value of a , the value of b and the value of c .

(4 marks)

A third function is $h(x) = 2 \cos\left(x + \frac{\pi}{3}\right)$.

- (b) State the range of $y = -(h(x) + 1)$.

(2 marks)

Supplementary page

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